

INTEGRATING ERC-20 ALLOWANCE EDGES INTO PAGERANK FOR ENHANCED ON-CHAIN WALLET REPUTATION SCORING

CHAPTER 5: Discussion

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5 Discussion

Chapter 4 reported what was measured on Arbitrum One. Each research question and each objective of Chapter 1 is judged against the criterion stated there. Each interpretation names the table of Chapter 4 it rests on. Same-window statements refer to the matched cohort and holdout statements to the spender cohort unless noted.

5.1 Findings by research question

5.1.1 RQ1: Reputation structure and rank divergence

RQ1 asks whether EndorseRank and AWP produce meaningfully different wallet orderings on an identical sample. The inter-method Kendall τ and its interval (Table 4.3, last row) sit well inside the open interval between unrelated and identical orderings: correlation is present, because both methods draw on overlapping wallet activity in one window, but a large share of pairwise orderings disagree (Figure 4.4).

The divergence is structurally expected. The endorsement graph encodes current spending authorization with about one ninth as many edges as the transfer graph, which encodes time-decayed economic flow^[3]. PageRank propagates influence along whatever edges it is given under the same damping convention^[15], and the sensitivity of the stationary vector to the edge set is well characterized^[23]; different edge sets therefore yield different stationary mass. A wallet that receives large transfers without granting allowances ranks highly under AWP and modestly under EndorseRank, and conversely for a wallet that authorizes spenders without being a transfer hub. H1 is supported. The practical corollary is that the two scores are not substitutes: adopting EndorseRank for efficiency means accepting a different ranking geometry, not a faster approximation of AWP.

5.1.2 RQ2: Construct-specific alignment

RQ2 asks which method aligns more strongly with contemporaneous allowance checks and with contemporaneous transfer checks. In addition to these overall results, Tables 4.3 and 4.4 display construct-specific winners. AWP wins the transfer family, while EndorseRank wins the allowance family, and the between-method paired differences exclude zero in opposite directions.

Two caveats apply to the reading of H2. First, there is a degree of methodological circularity in measuring each method’s agreement with its own construct: since the proxies in a family are all computed using the same event stream as the corresponding graph, the construct validity observed in RQ2 is more in the sense of Cronbach and Meehl^[22] than external validation. Second, there is no significant within-method contrast: EndorseRank’s alignment with allowance is not significantly higher than its alignment with transfer, and EndorseRank’s agreement with in-approve degree is not significantly higher than its agreement with transfer in-degree (Table 4.4, contrasts i and v). Thus, the evidence supports only the between-method statement (centrality of transfer does not reflect authorization and the centrality of endorsement does not reflect transfer volume as well as does AWP), and does not support the claim that EndorseRank is more about allowances than transfers in its own ranking. H2 is supported, in its construct-specific, between-method form.

5.1.3 RQ3: Computational efficiency

RQ3 asks whether EndorseRank is more computationally efficient than AWP on the same matched sample. Table 4.1 answers in the affirmative, in terms of wall-clock time, memory and number of iterations. The reported runtimes include the edge counts alongside them so that the ratio can be attributed to those counts: the endorsement graph is roughly one-ninth the size of the transfer graph, and the per-iteration cost is linear in $|E|$ ^[23]. This is because EndorseRank encodes authorization as a snapshot of the latest allowance, rather than an accumulated, decayed history of transfers^[3]; there is no difference in the computation between the two methods.

Table 4.2 provides a caveat rather than an extension of this result: the ratio of wall-clock times

stays within a narrow range over the different staged node counts tested, but the stages add address nodes without adding any extracted edges, and thus do not exercise how either method will scale up when the edge population increases. The results in Tables 4.8 and 4.9 indicate that changing the parameters of damping and top-token selection have no impact on the pattern of alignment; however, they do change the ratio of run-times. H3 is supported for the matched cohort, and for the limited experiments performed on node count.

5.1.4 RQ4: Temporal holdout of future new approvals

RQ4 consists of two clauses. The first asks whether scores computed up until the end of February are better predictors of new owner–spender approvals during the following three months, for inbound spenders, than AWP is. Table 4.13 indicates that EndorseRank is, but AWP is not; EndorseRank’s confidence interval excludes zero, and the between-method contrast is the strongest of the entire thesis (Table 4.14). Since the labels are all after the scored event, this is not a same-window construct check and so is the best evidence that endorsement centrality contains information about spenders that transfer centrality does not. The ordering is the same, though weaker, on the flow label: EndorseRank is better than AWP at predicting new transfer senders among this cohort.

The second part of the question asks whether EndorseRank is better than the same-window allowance degree, and it isn’t. The spender’s in-approve degree at t_1 predicts new approval pairs with a $\tau = 0.711$, EndorseRank with a $\tau = 0.479$, and the paired difference is -0.233 , with a confidence interval far below zero. AWP has the same relationship to the transfer in-degree on new senders. In other words, the number of new counterparties within a quarter is correlated with the number of old counterparties, and the local degree contains that information without the interference from owner-side centrality and allowance value caused by propagating the count. So the out-of-window portion of EndorseRank is derived from the degree it smooths, not in addition to it. This doesn’t negate the first half: EndorseRank still differs from AWP out of window by construct. But it eliminates the possibility that propagation provides predictive power beyond the degree for

this label, and the result holds equally for the transfer-graph baseline.

RQ4 departs from the research proposal in that we did not compare against a trading-success reference. We dropped this comparison because the outcome-native reference was based on the same GMX events as the proxies it would be compared against (Section 2.8); instead we compare against out-of-window but same-construct labels, i.e. new approvals and new senders, without any element of profit or repayment. What RQ4 demonstrates is the temporal predictive validity of the endorsement construct as compared to the transfer construct, and bounded above by the raw degree. It doesn't speak to credit risk.

5.1.5 RQ5: Coupling the two edge types

RQ5 asks whether a combined walk on both layers can rank wallets better than either individual layer. It cannot, given the setting fixed before running. On the holdout, C-PR at $\lambda = 0.5$ sits between EndorseRank and AWP on both labels, the paired difference to EndorseRank on new approval pairs is negative, and the paired difference to AWP on new senders includes zero (Table 4.14). The sensitivities also order monotonically, with higher allowance weights increasing τ on both labels, so that the transfer layer provides no additional predictive power out of window for spenders beyond what the allowance layer already provides. C-PR also has a secondary rule that we can test: in the same window (Section 4.7) C-PR tracks AWP to within a few thousandths on the transfer and Sybil-stability families and is indistinguishable from zero on the allowance family, so the secondary rule fails on its allowance part; the seeded ablation is below the method it borrows from on every family. The mixture is dominated by the denser layer, and λ acts only at the minority of nodes that have out-edges in both.

The ablation locates the reason. S-PR, which keeps the transfer walk and restarts it from the EndorseRank distribution, reproduces AWP on the flow label and loses almost the whole endorsement signal on the approval label. Once transfer edges govern propagation, an endorsement-derived teleportation vector changes little. The allowance edge earns its result by being walked on, not by being used as a seed, and mixing it with transfer edges at the transition level dilutes it in proportion to

the transfer weight. That is a statement about these two layers on this chain; a learned or node-dependent λ , or a coupling that keeps the layers separate and combines their stationary vectors afterwards, was not tested and is left to future work (Chapter 6).

We report this negative answer as a contribution rather than hide it, for two reasons. The rule was pre-registered, so the result cannot have been shaped to the data. And it bounds a natural next step for anyone who reads the crossed pattern of Table 4.3 as an invitation to combine: on this evidence a convex mixture of the two edge types at the transition level does not produce a score that beats the better single-layer method on either construct.

5.2 Achievement of the research objectives

Chapter 1 stated five objectives with a measurement and a criterion each. Table 5.1 judges them against the evidence.

Table 5.1: Research objectives, the criterion stated in Chapter 1, and the outcome.

Obj.	Criterion	Evidence	Outcome
O1	95% interval of inter-method τ excludes 0 and 1	Table 4.3, last row: interval lies strictly inside (0, 1)	Met
O2	Each method’s own-construct $\Delta\tau$ against the other method is positive with an interval excluding 0	Table 4.4, contrasts ii (AWP on transfer) and iii (EndorseRank on allowance)	Met
O3	Runtime ratio reported together with the edge ratio under one fixed protocol	Tables 4.1, 4.2 and 8.1: mean, SD, memory, iterations, $ V $, $ E $ per configuration	Met
O4	EndorseRank holdout interval excludes 0 and paired difference to AWP lies above 0; increment over t_1 in-approve degree reported with its interval	Tables 4.13 and 4.14: first part met; increment over degree is negative with an interval below 0	Met (first clause); not met (second clause)
O5	C-PR no worse than EndorseRank on new approval pairs and better than AWP on new senders; same-window position within the leading single method’s interval	Table 4.14: both holdout parts fail; Table 4.12 for the same window	Not met

Three findings that were not objectives also influence the conclusions. The within-method compar-

isons of Table 4.4 were not statistically significant, thereby restricting the inference space regarding EndorseRank’s class of methods (Section 5.1.2); the expanded-pool experiment tested node count, not edge growth (Section 5.1.3); and the raw t_1 degrees predict better than either PageRank on their own future-tense labels (Section 5.1.4). These are all reported as limitations on the claims, not incorporated into the objectives.

5.3 Theoretical implications

The results have implications for three areas of theory: propagation semantics on financial graphs, construct validity of on-chain scores, and the economics of trust under pseudonymity.

5.3.1 Propagation semantics beyond transfers

PageRank was originally formulated as a citation network on hyperlinks ^[15]. Do and Do^[4] extended the metaphor to Ethereum transactions, and Do, Do and Nguyen^[3] extended further to AWP by weighting transfer edges by value and a logistic decay on age. EndorseRank extends the metaphor to ERC-20 allowances: an owner who allows a spender to act on his behalf effectively issues a directed endorsement of spending authority ^[18]. The divergence detected under RQ1 demonstrates that altering edge semantics alters the ranking space even with the same nodes. Graph-based reputation is thus a family of methods, valid depending on the relationships encoded as edges ^[23]. Peer-to-peer and web-based trust propagation systems reached the same conclusion from the other direction: in EigenTrust ^[16] and TrustRank ^[17], outcomes are dominated by seed set and edge definitions. EndorseRank provides a new edge definition that is native to a token standard, not to a message-passing overlay or a hyperlink graph.

5.3.2 Multi-construct domain alignment

The check design isolates transfer centrality from allowance delegation and tests each method against both. The result, each method aligning more closely with its own construct and less so with the other’s, is what classical construct-validity theory would predict for measures that measure dif-

ferent constructs ^[22], and Messick’s ^[33] integrated view extends this to say that the consequences of employing a score are also part of its validity. Rank correlations ^{[20],[21]} operationalize this logic for ordinal reputation scores, without requiring classification thresholds, and the paired bootstrap ^[34] renders the pattern as testable. Instead, we suggest formal methods of comparison rather than informal comparison of point estimates. The null within-method contrasts demonstrate that convergent evidence on a method’s own construct is not equivalent to discriminant evidence within the method.

5.3.3 Reputation under cheap pseudonyms

Public ledgers allow cheap identity creation ^[5], and Ethereum’s account model makes address creation free of any registration step ^[36]. Reputation systems that reward inbound transfers or allowances are thus vulnerable to Sybil strategies, where the attacker creates edges among controlled addresses ^[11]. The empirical findings do not rule out this possibility; they demonstrate that the allowance and transfer graphs react differently to the activity observed in the observational sample. We formalize attack cost on allowance graphs in Section 5.6, arguing that non-zero allowances to external spenders create opportunity and revocation costs that transfer wash trading does not.

5.4 Practical implications and deployment scenarios

Protocols and infrastructure providers face a design choice between transfer-hub detection and authorization-aware reputation. The empirical trade-off supports scenario-dependent adoption rather than a single recommended score.

5.4.1 Transfer-hub and flow monitoring

Protocols that prioritize detecting high-flow wallets (for market-maker identification, routing analysis, or transfer-based compliance heuristics) have reason to prefer AWP despite its cost. Its transfer alignment (Table 4.5) makes it the better fit when the operational target is economic flow centrality^[3]. Its runtime on the matched cohort (Table 4.1) is acceptable for batch analytics and

less attractive for near real-time refresh.

5.4.2 Authorisation-aware and low-latency refresh

Protocols that emphasize explicit spending authorization (token vaults, allowance-gated integrations, or risk engines that view approvals as trust acts) have reason to prefer EndorseRank. Its allowance alignment (Table 4.6), its holdout result (Table 4.13), and its sparse graph and short solve time (Table 4.1) make it well-suited to frequent recomputation. Operators can tolerate weaker transfer alignment in exchange for authorization semantics, out-of-window evidence, and speed. One caveat follows from Table 4.14: an operator whose sole question is “How many new approvals will a spender attract?” is better served by the spender’s current in-approve degree than by any PageRank. The propagated score should be used only for what it adds, which is resistance to the weight of a single owner and a ranking that accounts for who the endorsing owners are.

5.4.3 Evaluation design and combined use

Combining the two signals is both a deployment question and a method question, and the evidence answers each differently. It is possible to run EndorseRank for continuous authorization-aware monitoring and AWP for periodic transfer-hub audits; the cost is the sum of the two solves. However, folding the two edge types into one walk is not supported: the coupled operator costs as much as AWP (Table 4.1) and ranks no better than the appropriate single-layer method on either construct (Section 5.1.5). If an operator wants both constructs, they should compute both scores and keep them separate. The open pipeline allows cross-protocol replication on any ERC-20 chain with allowance logs (Appendix 8.1). The integration with live risk oracles is left as future work (Chapter 6), though EndorseRank’s speed could support a faster refresh cycle than the transfer-graph method would allow without incurring equivalent costs.

5.4.4 Industry adoption scenarios

Both approaches provide a cheap, standardized and independently recomputable ordinal signal. Value capture is hypothesized to come from fees, utilization, monitoring cost and reduced reward leakage rather than collateral. The scenarios below describe actors facing decisions that are addressed by the results.

- DEX operators. Protocols earning from trading fees face the problem of which wallets to prioritize for monitoring, fee tiers or feature access when they update their reputation frequently. EndorseRank is suited to this workflow because its score reflects authorization activity and can be recomputed in seconds, avoiding the need to aggregate transfer history. Any revenue gain would flow through fee volume and reduced monitoring cost; therefore, fee and monitoring policies represent the appropriate lever.
- Lending protocols. Lending markets want greater capital utilization without reducing collateral ^[41]; the stress events of 2020 demonstrate the consequences of delayed risk parameter updates ^[9]. Reputation can serve as an additional signal for interest rates or fee tiers and for reviewing collateralized positions, not as collateral replacement. EndorseRank is the appropriate input when a lending protocol already uses ERC-20 approvals for deposits and borrowing (Section 2.7). The scores are not default prediction tools; experiments with different collateral requirements are left to future work.
- Wallet and security tooling. Users and integrators want to know if a spender contract is broadly authorized. Showing a spender's EndorseRank, or the total incoming endorsement mass, as a trust badge turns allowance semantics into explicit delegation ^{[18],[19]}. The badge communicates the position of a graph node under a defined set of edges; it is not a security audit or a guarantee against malicious contracts.
- Liquidity providers and market makers. These participants need to identify transfer hubs and flow structure for inventory and spread control. The AWP approach is more suitable for that goal^[3]; the benefit is reduced screening cost, not higher profit and loss.

- Airdrop and governance programmes. Token distributions and governance eligibility are routinely abused by Sybil clusters ^{[11],[32]}. Programs can mandate diversity of incoming endorsements or use the closed-cluster mass limit from Section 5.6 (Equation 5.8) as a gating mechanism before distributing rewards. Such conditions reduce the impact of closed-cluster inflation; they cannot prevent open-cluster attacks recruiting honest incoming endorsements.
- Analytics and compliance providers. Exchanges, funds and compliance teams require auditable counterparty rankings that can be recomputed externally. The open source codebase facilitates the publication of EndorseRank and AWP scores using defined windows and parameters; the distinguishing feature is auditability compared to proprietary machine learning models ^[2]. Providers should disclose Sybil exposure and that the scores are not profit predictions alongside any published ranking.

In no case does reputation replace collateral; the standardized, auditable and low-latency ordinal signal serves to decrease the expense of counterparty differentiation.

5.4.5 Standardised scores as shared infrastructure

The point of any standardized financial metric, such as a reputation score, is that it is public, comparable and cheap to recompute, so that multiple parties can make judgments based on the same number, and understand how the metric may be manipulated. Its predictive capability is less important. EndorseRank and AWP fulfill the criteria of an infrastructure-level system for pseudonymous wallets because they are computed from public ledgers using an open-ended pipeline (Appendix 8.1), can be refreshed quickly, and are reproducible from the public ledger. They are not intended as forecasts of profitability, and neither can be useful unless their computation is public and their vulnerabilities to manipulation are specified (Sections 5.6 and 5.7).

5.5 Societal and practical impact

Counterparty vetting is expensive and uneven in the DeFi market. Wealthy actors may afford proprietary risk analysis tools that analyze entire transfer histories; retail participants rely on anecdotal evidence or do not vet at all. A tool that can be updated at a fraction of the cost of re-analyzing the entire transfer graph (Table 4.1) and that leverages the permissions that users have already granted, decreases the barrier to entry to counterparty vetting. Furthermore, because the mechanism used in most token draining exploits relies on approving access to funds (Section 2.17), a spender-centered endorsement score provides information about a risk that users face directly, and that transfer volume cannot measure.

Protocol developers are provided with an argument for replacing a singular “reputation” construct with specific scoring mechanisms tailored to individual decisions. This change will mitigate the risk of incorrect application (using a flow score to determine authorization, for example) and, since both metrics can be recalculated from public data through a transparent pipeline, allow users themselves to audit their exposure. The reduction in computational cost will also make continuous monitoring affordable for protocols that cannot afford to periodically recalculate the transfer graph at the same expense.

For society at large, this paper demonstrates that a useful on-chain reputation metric can be constructed in an entirely transparent manner: the input events are public, the parameters are versioned, and each of the tables can be recreated from scratch based on those inputs. Unlike the opaque decentralized credit scoring systems discussed by Packin and Lev-Aretz^[2], this system provides the type of explanatory information that regulators suggest should be required when analyzing DeFi protocols^[63]. However, it also introduces a duty, described in Section 5.7, to ensure that the ordinal nature of a graph-based statistic is not mistaken for an identity, credit history, or proof of personhood.

Since these experiments were conducted on a single blockchain over a six month period, the results relate to the construction of endorsements rather than credit risk. To produce social value, therefore,

these results must be deployed within the boundaries of their construction. Every reputation metric that impacts the allocation of value invites the possibility of exploitation. While the following section details why the costs involved in manipulating these metrics are high, there is no proof that the manipulation is impossible.

5.6 Sybil attack cost model on allowance graphs

A Sybil attack relies on inexpensive pseudonyms to generate a false sense of trustworthiness ^[11]. In a transfer graph, a malicious actor can artificially boost in-degree and in-flow by moving assets between accounts under their control, at the expense of transaction fees and potential inventory loss ^[29]. On allowance graphs, the attack surface is different, as edges reflect authorizations instead of actual transfers. In this section, we formulate a simple cost model for inflating EndorseRank using fake allowances.

5.6.1 Attack setup

Suppose the attacker controls a set of addresses $\mathcal{A}$ and let $\mathcal{H}$ be the set of honest wallets in the endorsement graph $G_{\text{approve}} = (V, E, w)$. An edge (u, v) with weight $w_{uv} > 0$ means that u has authorized v to spend w_{uv} tokens (after decimal adjustment). EndorseRank solves for the PageRank vector $\mathbf{r}$ on G_{approve} with damping parameter $d \in (0, 1)$ ^[15]:

$$\mathbf{r} = d P^\top \mathbf{r} + \frac{1-d}{|V|} \mathbf{1}, \quad (5.1)$$

where P is the row-stochastic matrix derived by normalizing outgoing allowance weights, with the standard treatment of dangling nodes ^[23]. The attacker aims to maximize the average EndorseRank of a target set $T \subseteq \mathcal{A}$,

$$J(T) = \frac{1}{|T|} \sum_{i \in T} r_i, \quad (5.2)$$

subject to a budget constraint on the cost of attack, which we define next.

5.6.2 Cost components

The creation of each allowance edge (u, v) with weight w_{uv} entails at least the following costs.

1. Gas cost. Calling approve or permit requires gas. Let c_{gas} be the expected fee per allowance update. With m updates,

$$C_{\text{gas}} = m c_{\text{gas}}. \quad (5.3)$$

2. Capital lock and opportunity cost. An open allowance to an external spender $v \notin \mathcal{A}$ means that the tokens under u can be transferred by v . Even when $v \in \mathcal{A}$, large allowances to certain contracts or routers may be necessary to appear legitimate. Let $\ell(w_{uv})$ be the opportunity cost of capital tied to the edge with weight w_{uv} during the period covered by a reputation refresh (e.g., forgone interest). Then

$$C_{\text{cap}} = \sum_{(u,v) \in E_{\mathcal{A}}} \ell(w_{uv}), \quad (5.4)$$

where $E_{\mathcal{A}}$ is the set of edges created by the attacker.

3. Revocation and detection cost. Legitimate users periodically revoke old allowances. Attackers who do not revoke might be identifiable. Let c_{rev} be the expected cost of revoking and approving again periodically in order to maintain a desired set of edges for k reputation refresh periods:

$$C_{\text{rev}} = k c_{\text{rev}}. \quad (5.5)$$

The total cost of the attack is given by

$$C = C_{\text{gas}} + C_{\text{cap}} + C_{\text{rev}}. \quad (5.6)$$

5.6.3 Influence and a cost-effectiveness bound

Under common sensitivity results for PageRank, the effect of an edge on $\mathbf{r}$ is proportional to the stationary mass at the source node and the normalized outgoing edge weight ^[23]. As a simple

bound, consider the impact of adding edges only among the attacker’s own nodes (i.e. a closed Sybil network with no incoming endorsements from honest nodes). The maximum increase in $J(T)$ from adding edges among $\mathcal{A}$ alone is limited by the teleportation mass:

$$\sum_{i \in \mathcal{A}} r_i \leq |\mathcal{A}| \cdot \frac{1-d}{|V|} + d \cdot \sum_{i \in \mathcal{A}} r_i, \quad (5.7)$$

which rearranges to

$$\sum_{i \in \mathcal{A}} r_i \leq \frac{|\mathcal{A}|}{|V|}. \quad (5.8)$$

Hence, a purely closed attacker cluster cannot capture more than its population share of total rank mass. To beat this bound, the attacker must obtain incoming endorsements from $\mathcal{H}$. Doing so generally requires either stealing the private keys of legitimate wallets or persuading them to authorize the attacker’s accounts; both involve costs that are dominated by C_{cap} and social-engineering effort, and not by the transaction fees themselves.

When the attacker buys inbound allowances from legitimate owners (e.g., by running an app that asks users for permission), the marginal cost of one unit of honest-sourced weight w_{ha} may be expressed

$$c_{\text{in}}(w_{ha}) = c_{\text{gas}} + \ell_{\text{honest}}(w_{ha}) + \pi(w_{ha}), \quad (5.9)$$

where ℓ_{honest} is payment made to the honest owner and π is a risk premium in case the allowance is spent by a bad actor. Then the cost-effectiveness of the attack is

$$\eta = \frac{\Delta J(T)}{C}, \quad (5.10)$$

where $\Delta J(T)$ is the increase in mean rank of the target from the attack edges.

Equations (5.7)–(5.10) show that EndorseRank is not Sybil-proof, but that the amount of closed-cluster inflation is bounded by population share and that cross-cluster inflation requires purchasing honest-sourced allowances at cost. Wash trading transfers have other costs (fees and inventory)

but need not expose spending permissions to outside accounts. The Sybil-robust stability proxies of Chapter 4 mitigate some forms of observational manipulation; however, they do not supplant adversarial assessment, which we leave as future work.

5.7 Regulatory and ethical considerations

On-chain reputation metrics pose questions regarding both transparency and equity. Prior legal work has established that decentralized finance markets must still comply with disclosure and consumer protection regulations despite on-chain settlement ^[63]. Packin and Lev-Aretz^[2] also warn that decentralized credit scoring may perpetuate black-box harms if models are uninterpretable, uncontestable, or linked to identity databases without due process. Both EndorseRank and AWP are deterministically computed from open-source algorithms and on-chain events, allowing anyone to recalculate them using the open-source software (Appendix 8.1). This provides a degree of openness against black-box criticism, but does not alleviate the concern of distributional impact, as addresses with limited transaction activity will receive lower scores independent of off-chain creditworthiness.

In terms of ethics, these scores should never be used as proxies for identity, as measures of creditworthiness in the traditional sense, or as evidence of a unique individual ^[11]. Instead, they are rank-based indicators of network position within the defined graph topology. Any deployment that relies on score thresholds to grant financial services should clearly document the underlying graph definitions, timeframe, and limitations, particularly with respect to Sybil vulnerability. Regulatory frameworks focused on interpretability would prefer this method of ranking to the use of proprietary machine learning models like those reviewed by Kotb ^[14], so long as providers do not make exaggerated claims about their ability to predict loan repayment rates ^[2].

Privacy is inherently constrained on open blockchains ^[5]; the process of calculating reputation scores relies only on publicly available information. Consequently, the main ethical consideration is the manner in which the results are utilized; additional attempts to reveal the identities behind each

address are discouraged, and the scores should not be utilized as permanent identifiers comparable to fingerprints. The current study focuses exclusively on wallet addresses and does not attempt to connect them to individuals in the real world (Section 3.16).

5.8 Threats to validity

The findings reported here should be evaluated in light of the following limitations, categorized according to construct, internal, external and adversarial validity.

5.8.1 Construct validity

- Ratings are not credit ratings. No default labels associated with uncollateralized lending are used.
- Same-window checks are allowance and transfer local statistics; the holdout labels are future new approvals and future new transfer senders.
- Holdout construct. Future new approvals are still endorsement events and future new senders are still flow events. They are out of the score window, but they are not an independent credit or trading construct, and both are counts of arrivals, which favors any score that tracks the current count of counterparties (Section 5.1.4).
- Own-construct agreement is partly mechanical. EndorseRank is a global smoothing of spender in-approve degree, and AWP of decayed transfer in-strength; each method's agreement with its own family is expected by construction, which is why the within-method contrasts of Table 4.4 are reported and why the null result on them is treated as informative.

5.8.2 Internal validity

- Shared observation window. All methods and proxies are computed over 1 December 2025 to 31 May 2026. Market regimes specific to that window (volatility, incentive programs, protocol changes) may inflate or deflate alignment. Robustness checks on damping and top-

token subgraphs reduce sensitivity to hyperparameters and token concentration but do not vary the calendar window.

- Cohort selection. The matched cohort requires non-zero connectivity in both reputation subgraphs. Results may not extend to addresses absent from allowance and transfer subgraphs; the sample-definition sensitivity of Table 4.10 shows how strongly the coefficients respond when that requirement is relaxed.
- Edge-weight measurement. Token-decimal normalization, unit conversion and permit-versus-approve coverage can affect w_{uv} . Measurement error in edge weights propagates to $\mathbf{r}$ and to rank correlations [23].
- Timing environment. Runtimes are wall-clock measurements on one machine; the standard deviations in Table 4.1 bound within-session noise but not between-machine variation, which is why ratios rather than seconds are the comparative summary.

5.8.3 External validity

- Single chain and window. Evidence is limited to Arbitrum One over six months. Generalization to other rollups, Layer-1 chains or lending-centered venues requires cross-protocol validation; DeFi venues differ in protocol design [8] and in market structure [1].
- Pool scaling versus alignment scaling. The expanded pool supports a node-count statement about runtime only (Section 4.2.1); same-window alignment remains anchored to the matched cohort, and the holdout to spenders. Efficiency gains should not be assumed to preserve the same τ profile at arbitrary population sizes.

5.8.4 Adversarial validity

- Sybil and wash trading. Adversarial allowance or transfer patterns may manipulate rankings [11]. Sybil-adjusted proxies partially address but do not eliminate this risk. The cost model of Section 5.6 is analytical; it is not an empirical red-team evaluation.
- Strategic approval behavior. Once EndorseRank is deployed, rational actors may grant al-

allowances strategically to farm rank. Observational τ estimated before deployment may overstate robustness to strategic behavior.

5.9 Limitations

Results are based on a single six month period on Arbitrum One for a matched wallet cohort and a spender holdout cohort, the number of users and graph size of which are reported in Chapter 4. EndorseRank’s strengths include construct alignment with allowances, its rank being distinct from AWP, computational efficiency due to graph sparsity, and its holdout being aligned with future new approvers whereas AWP shows no such alignment; its weaknesses include less construct alignment with transfers than AWP, less construct alignment with longitudinal stability than AWP, an allowance lead relative to its own transfer alignment that was not significant, and a holdout coefficient lower than that of the raw in-approve degree it smooths.

The coupled operator was evaluated at three fixed values of λ with uniform teleportation and a per-node convex mixture at the transition level; node-dependent or learned weights, mixtures of the stationary vectors instead of the transition rows, and other Markov chains were not tested, hence the negative answer to RQ5 is limited to that family of methods on this dataset. Normalization by token decimals, currency conversions, and permit versus approve coverage could influence edge weights. The expanded-pool experiment evaluates node growth, not edge growth, since the second tier extraction did not finish. Longer-running baseline runs (other PageRank variants and a graph constructed around lending activity) have been stored for future comparison (Appendix 8.4) but are omitted from the main comparison. The dataset does not contain default labels for unsecured lending. Even if results are construct-aligned, regulatory and ethical considerations regarding their use (Section 5.7) restrict their deployment.

These limitations constrain the conclusions to be drawn: EndorseRank is a viable, computationally efficient, allowance-specific social reputation method, for this cohort and time frame; it is not a credit scoring method, nor a Sybil-resistant identity mechanism, nor a substitute for protocol-level

collateralization ^[9]. Chapter 6 repeats what has been shown, the value of that knowledge and what would be required to extend these bounds.